Priya Patel

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Education

Master of Science in Data Science | SUNY, University at Buffalo GPA: 4.00 Jun 2024
Master of Science in Applied Physics | The M. S. University of Baroda GPA: 4.00 Jun 2019

Skills

Programming: Python *(Scikit-learn, Pandas, NumPy, PyTorch, TensorFlow)*, MATLAB, R, SQL, PostgreSQL, Shell
Softwares and Tools: Power BI, Tableau, GCP BigQuery, Cloud SQL, GCP Looker, Git, Microsoft Excel, Microsoft PowerPoint
Data Science Skills: Data Visualization, Predictive Modeling, A/B Testing, High-Performance Computing, Database Management, Data Integration, Statistical Analysis, CRM, Healthcare Data Analysis, Medicare Data, Survival Analysis
Cloud Environments: Google Cloud, Microsoft Azure, AWS, Snowflake

Experience

Business Intelligence Analyst | BestSelf Behavioral Health, Buffalo Oct 2024 - Present

- Implemented automated monthly reports for major depressive disorder outpatient treatment tracking using SQL, RT Report Manager, and Crystal Reports, ensuring complete patient information for medical directors.
- Developed a denial write-offs dashboard in Power BI to track denial trends, payer details, and financial impacts, helping leadership identify patterns and reduce claim denials to improve revenue flow by 12%.
- Built a real-time dashboard for counselors and program directors of 69 programs across 13 clinics to track high-risk suicidal clients, enabling proactive intervention and reducing missed appointments by 27%.
- Created comprehensive reports for Erie County and Renaissance Addiction Services, tracking unique clients served, demographics, service utilization, and program engagement to support better resource allocation and policy decisions.
- Developed automated Power BI reports with RT report manager to identify medication management appointments outside the two-week compliance window, ensuring adherence to new state protocols and minimizing scheduling errors.
- Designed a compliance dashboard for legal and compliance teams to assess HIPAA violations, security incidents, and treatment plan adherence, providing data-driven insights for risk mitigation and policy improvements.
- Collaborated with cross-functional teams, including senior leadership, legal, compliance, billing, and program directors, to generate and track custom Key Performance Indicators (KPI) for operational improvements.

Data Analyst (Graduate Research Assistant) | University at Buffalo, Buffalo Aug 2023 - Sep 2024

- Collected and analyzed portfolio data using Fama-French factors to perform a A/B testing of two investment approaches: investing every month throughout the year and investing in all months except the summer months.
- Employed a Monte Carlo simulation approach to evaluate the robustness by randomly selecting 30-year investment periods.
- Performed Statistical Analysis to calculate key performance metrics, including the Time-weighted rate of return and Sharpe ratio, to provide insights into risk-adjusted portfolios.

Graduate Teaching Assistant | University at Buffalo, Buffalo Jun 2023 - Jun 2024

- Provided mentorship to 250+ students for Python for Data Science, focusing on applications of machine learning in industry.
- Designed and conducted hands-on workshops on deep learning and neural networks using TensorFlow, enhancing students' understanding of real-world ML applications.

Financial Data Analyst | Dark Horse Stocks, India Jan 2022 - Jan 2023

- Utilized SQL and Python to automate data collection on AWS server from BSE and NSE, reducing manual data entry time.
- Performed time series analysis using statistical models to forecast portfolio performance, and evaluate the impact of investment on various financial scenarios to enhance clients' understanding and confidence in investment choices.
- Maintained dynamic Power BI dashboard to produce weekly market insights and investment recommendations, contributing to an 8% increase in client engagement.

Data Scientist (Research Assistant Professor) | Saurashtra University, India Jun 2019 - Jan 2022

- Utilized Alteryx's data-wrangling, data-cleaning, and modeling techniques to create a new database by merging data from various government databases using SQL for assessing female education policies impacting over 3,00,000 students.
- Provided data-driven insights to policymakers using hypothesis testing for strategically restructuring the education budget that prioritizes women in need, identified using Spectral Clustering algorithm.
- Delivered lectures for a class of Statistics, Quantum Mechanics & Classical Mechanics for final-year bachelor students.

Data Science & Machine Learning Projects

Network Based Data Analysis in Omics Data | University at Buffalo Aug 2023 - June 2024

- Working on applying advanced bioinformatics methods to address challenges in the analysis of high-dimensional omics data (8000 cells $\times$ 21000 gene expressions) using AnnData and Scanpy libraries along with Numpy, Pandas and Scikit-learn.
- Employing latent representational learning to reconstruct reduced spaces capturing underlying data structures.
- Utilizing SCREAM method for clustering single-cell Multiomics data obtained from combined scRNA-seq and scATAC-seq.

Chronic Kidney Disease: A Data-Driven Diagnosis | University at Buffalo Jan 2023 - May 2023

- Preprocessed the raw data by handling missing values, feature reduction, and data normalization to ensure model performance.
- Implemented exploratory data analysis to gain insights into the dataset, identify trends & outliers, and select relevant features.
- Developed and implemented multiple statistical learning models, including logistic regression, support vector machines (SVM), decision trees, random forests, and neural networks, using Python and libraries such as scikit-learn and TensorFlow.
- Conducted a comparative analysis of different models' results, assessing performance and suitability for diagnosis.

Market Trend Analysis and Forecasting Using Statistical Models | University at Buffalo Aug 2023 - Dec 2023

- Conducted analysis of financial markets using SQL and Tableau to identify trends and economic indicators.
- Developed predictive models using techniques such as LSTM (Long Short-Term Memory) and ARIMA to forecast market trends and movements. And evaluated the models using metrics such as accuracy, precision, recall and F1-score.
- Implemented feature engineering strategies to enhance model performance by 23%.

Heart Bounding Box Prediction from X-Ray Images | University at Buffalo Jan 2024 - Jun 2024

- Developed a CNN model based on modified the ResNet-18 architecture that accommodates grayscale X-ray images as input and predicts four coordinates of bounding box corners, utilizing a dataset of 469 annotated images.
- Implemented various preprocessing and data augmentation techniques like random contrast changes, scaling, rotations, and translations to increase the size of the dataset, resulting in model robustness.
- Optimized computational efficiency by leveraging transfer learning with pre-trained weights for all layers except the first and last layers, thereby fine-tuning the model for heart detection.
- Attained an impressive prediction accuracy of approximately 95%, depicting the efficacy and reliability of the model.

Airport Database Management in SQL | University at Buffalo Aug 2023 - Dec 2023

- Handled data normalization utilizing BCNF analysis to enhance the structure and memory efficiency of the database.
- Executed multiple queries using PostgreSQL to extract valuable insights, contributing to informed decision-making processes.
- Adapted query optimization strategies to reduce computation time by 41%.
- Designed a interactive Power BI dashboard to visualize key metrics and trends for intuitive data exploration.

Certifications

Microsoft Certified: Azure Fundamentals
Deep Learning with PyTorch for Medical Image Analysis
Statistics with R Specialization
Data Visualization with Tableau
Database Management with PostgreSQL